Hardware-Software Interface

Introduction to the Course at OUC

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Overview

- Who am I?
- How the course works
  - Lectures and labs
  - Assessments
- What's different about the OUC version
- What you know already

About me

- Dr Adam SAMPSON – please call me Adam!
  Email: `a.sampson@hw.ac.uk`
- Associate Professor in Computer Science,
  at Heriot-Watt University's Edinburgh campus
- **Teaching:** systems programming, software security, networking – also MSc senior programme director
- **Research background:** concurrent programming languages, large-scale complex systems simulation
- Open source development

About me

- Unlike some other Heriot-Watt staff, I don't sound Scottish...
- I'm originally from Sevenoaks, in the county of Kent, near London
- … so I speak with a London-ish accent (sort-of a “BBC” accent)
- I've lived in Dundee since 2010

Course content

- C
  - C programming, using modern C standards
  - Basic embedded development with the Raspberry Pi
- ARM
  - 32-bit ARM assembler programming
  - Further embedded development: controlling devices
- Systems
  - Modern computer architecture and operating systems

Videos

- Lecture videos are available for most of the lectures
  - All of the C and ARM lectures have videos
  - Most of the systems lectures have videos
  - The tutorials and practicals usually aren't recorded

- If you miss a lecture owing to illness, please make sure you watch the video to catch up on what you've missed
- You can also use them for revision and reference later

Assessment

- There is **no exam** for F28HS

- Coursework 1 (40%) – Image Processing
- Coursework 2 (60%) – PIN Cracking
  - 6% of CW2 comes from an assessed lab exercise in Week 2

Coursework 1 – due 20th April

- Image processing: write a program to perform various operations on a picture
- Using C on Linux (week 2 material)
  - You can do this on any Linux machine, but I **strongly** recommend that you use your Raspberry Pi
- Individual coursework
- Worth 40% of your final mark for the course

Coursework 1 – due 20th April

- Coursework spec is available on Canvas (show it)
- Each student has a different list of tasks to do – make sure you've found your tasks, using the list on Canvas
- We will use the last few lab sessions while I'm here to work on the coursework – get feedback on your code
- You must demonstrate your work in progress to me on April 2nd-4th – I don't expect it to be complete!

Coursework 2 – due 20th May

- PIN cracking system, building software and hardware
- Pair coursework – two students work together, and both submit the same thing
- Tests C programming, assembler programming, and systems development on Raspberry Pi
- Worth 60% of your final mark for the course
  - 6% of this is for the traffic lights assessed lab exercise – we will do this in the labs in week 2

Coursework 2 – due 20th May

- Pair coursework
- Sign up for a pair **on Canvas** by the end of week 3
  - F28HS > People > OUC CW2 Groups (show it)
  - May need one group of 3 if there's an odd number of students
- You need to divide the work between yourselves in a fair way – the report will ask you to describe who did what

Coursework 2 – due 20th May

- Coursework spec is available on Canvas (show it)
- Code template is provided, and we will do some parts of it during the lab exercises

- **Difference for OUC:** We can't do in-person demos for CW2, so your submission must **include a video**, showing your system working
  - Video doesn't need to be neatly edited – phone video is fine

Coursework submission

- Coursework 1 and Coursework 2 work the same way

- Submission through Canvas
  - Make sure your .zip file includes everything the coursework spec asks for, **plus the video** for CW2
  - For CW2, **both** students must submit the **same** file
- If you cannot access Canvas – email the file to me instead!

Academic misconduct

- Everything you submit for CW1 and CW2 must be **your own work** – we're trying to assess **your** skills!
- Do not submit work copied from other students
- Do not submit work generated by an LLM
  - See the guidance about GenAI use in F28HS on Canvas
- If we have doubts about student misconduct when marking, we have to report it for investigation; several students had to resit last year because they copied work

Late submission and MC

- We will follow the Heriot-Watt rules for late submission
- If you submit up to 7 days after the deadline, your mark is multiplied by 0.7 – more than 7 days will not be marked
- If you are affected by circumstances beyond your control (e.g. illness), please submit **Mitigating Circumstances**
  - We can then remove a late penalty, or take other actions
  - There's **no downside** to submitting MC – it can only improve your mark, and OUC students often forget about it!

What's different for OUC?

- This year, F28HS is taught to 474 students across five campuses: Dubai, Edinburgh, Malaysia, OUC and Aktobe
- Differences:
  - Taught at OUC over 4 weeks, Edinburgh over 12 weeks – so the order of the material is different
  - Edinburgh uses Raspberry Pi 2 (“RPi2”) boards; OUC use newer Raspberry Pi 4 (“RPi4”) boards – some minor differences which I'll explain in the tutorials
  - In-person demo for CW1, video demo for CW2

Have you done C before at OUC?

- Many OUC students have an advantage in F28HS!
- Prof Qu/Prof Ye's material has covered the basics of C:
  - Lecture 1a: Introduction
  - Lecture 1b: Control Flow
  - Lecture 2a: Functions and Pointers
  - Lecture 2c: Arrays
  - Lecture 4a: Strings
- It's OK to skip those, but please attend the rest! Don't assume all the C content is material you've done before

Getting help

- I'm here to help you! I'll be in Qingdao until 4th April, and available by email or Teams call after that
- Please speak to me, or the other lecturers, if you need help

- If I say anything you can't understand, please tell me!
- The coursework specifications are quite complex – please read through them, and ask me if anything is unclear

Getting to know you

- Enough about me!

Put up your hand if…

- … you have done any C programming before?

  (Many OUC students have done C in a previous course)

Put up your hand if...

- … you'd consider yourself an **experienced** C programmer?

  (You could write a program that does something complicated without needing help)

Put up your hand if...

- … you've done any assembler programming before?

  (Which architecture? ARM, MIPS, x86, RISC-V...)

Put up your hand if...

- … you've used Linux before?

Put up your hand if...

- … you'd consider yourself **familiar** with Linux?

  (If I told you to download and compile a program, you'd be able to do it without more detailed instructions)

Put up your hand if...

- … you've done some electronics before?

  (Most people, I expect!)

Put up your hand if...

- … you've done some embedded programming before?

  (e.g. on a Raspberry Pi, Arduino or ESP8266 board)

Put up your hand if...

- … you've used a device with an I2C or SPI bus before?

Put up your hand if...

- … you've got any questions?

- I'll be looking for students to eat lunch or dinner with on the days that I'm teaching – if you'd like to talk to me about anything, then please come along!
  - Most common question last year was...